



GM12878 Micro-C

Pre-alignment

- Genome file is at `/scratch/users/kalyanna/ref/hg38.fa`
- Index file <faidx> is at `/scratch/users/kalyanna/ref/hg38.fa.fai`. To generate index file use command:

```
samtools faidx hg38.fasta
```

- Genome file contains first 2 columns of an index file and is stored at `/scratch/users/kalyanna/ref/hg38.genome`. To generate genome file run command:

```
cut -f1,2 hg38.fasta.fai > hg38.genome
```

- Bwa index file is stored at `/scratch/users/kalyanna/ref/hg38.fa.bwt`. To generate use command:

```
bwa index hg38.fasta
```

Alignment

Use bwa mem algorithm to align pairs:

```
bwa mem -5SP -T0 -t<threads> \  
<ref.fasta> <MicroC_R1.fastq> <MicroC_R2.fastq> \  
-o <aligned.sam>
```

- The aligned .sam file is at `/scratch/users/kalyanna/GM12878_micro_c/aligned_MicroC_800M.sam`
- **Recording valid ligation events.** We use the `parse` module of the `pairtools` pipeline to find ligation junctions in Micro-C (and other proximity ligation) libraries. When a ligation event is identified in the alignment file the pairtools pipeline will record the outer-most (5') aligned base pair and the strand of each one of the paired reads into `.pairsam` file (pairsam format captures SAM entries together with the Hi-C pair information). In addition, it will also assign a pair type for each event. e.g. if both reads aligned uniquely to only one region in the genome, the type UU (Unique-Unique) will be assigned to the pair. The following steps are necessary to identify the high quality valid pairs over low quality events (e.g. due to low mapping quality):

```
pairtools parse --min-mapq 40 --walks-policy 5unique \
--max-inter-align-gap 30 --nproc-in 8 --nproc-out 8 \
--chroms-path hg38.genome aligned_MicroC_800M.sam > ligation_MicroC_800M.pairsam
```

- **Sorting the pairsam file.** The pairsam pairs are then sorted using pairtools sort.

```
pairtools sort --nproc 16 \
--tmpdir=/scratch/users/kalyanna/GM12878_micro_c/tmp/ \
/scratch/users/kalyanna/GM12878_micro_c/ligation_MicroC_800M.pairsam
/scratch/users/kalyanna/GM12878_micro_c/sorted_MicroC_800M.pairsam
```

- **Remove PCR duplicates.** `pairtools dedup` detects molecules that could be formed via PCR duplication and tags them as “DD” pair type. These pairs should be excluded from downstream analysis. Use the pairtools dedup command with the `--output-stats` option to save the dup stats into a text file.

```
pairtools dedup --nproc-in 8 --nproc-out 8 --mark-dups \
--output-stats /scratch/users/kalyanna/GM12878_micro_c/stats_MicroC_800M.txt \
--output /scratch/users/kalyanna/GM12878_micro_c/dedup_MicroC_800M.pairsam \
/scratch/users/kalyanna/GM12878_micro_c/sorted_MicroC_800M.pairsam
```

- **Split the final .pairsam file into .sam and .pairs.** `.pairsam` has two extra columns containing the alignments from.

```
pairtools split --nproc-in 2 --nproc-out 2 \
--output-pairs mapped_pairs_MicroC_800M.pairs \
--output-sam unsorted_MicroC_800M.bam dedup_MicroC_800M.pairs
```

```
#!/bin/bash
#SBATCH -c 16
#SBATCH --mem=120G
#SBATCH --partition=altemose
#SBATCH --time=36:00:00
#SBATCH --chdir=/scratch/users/kalyanna/GM12878_micro_c/
#SBATCH --export=all
#SBATCH --requeue
#SBATCH --output=/scratch/users/kalyanna/GM12878_micro_c/sbatch_logs

source activate hic
```

```

pairtools dedup --nproc-in 8 --nproc-out 8 --mark-dups \
--output-stats /scratch/users/kalyanna/GM12878_micro_c/stats_MicroC_
--output /scratch/users/kalyanna/GM12878_micro_c/dedup_MicroC_800M.p
/scratch/users/kalyanna/GM12878_micro_c/sorted_MicroC_800M.pairsam

pairtools split --nproc-in 8 --nproc-out 8 \
--output-pairs /scratch/users/kalyanna/GM12878_micro_c/mapped_pairs_
--output-sam /scratch/users/kalyanna/GM12878_micro_c/unordered_MicroC_
/scratch/users/kalyanna/GM12878_micro_c/dedup_MicroC_800M.pairsam

samtools sort -@16 -T /scratch/users/kalyanna/GM12878_micro_c/tmp/ \
-o /scratch/users/kalyanna/GM12878_micro_c/mapped_MicroC_800M.PT.bam
/scratch/users/kalyanna/GM12878_micro_c/unordered_MicroC_800M.bam

samtools index /scratch/users/kalyanna/GM12878_micro_c/mapped_MicroC_

```

Process .pairs into .cool and .npz

The pairs are processed using cooler clod, and then converted to npz using [cool2npz.py](#) script:

```

#!/bin/bash
#SBATCH -c 16
#SBATCH --mem=32G
#SBATCH --partition=normal,altemose,owners
#SBATCH --time=48:00:00
#SBATCH --chdir=/scratch/users/kalyanna/GM12878_microc/
#SBATCH --export=all
#SBATCH --requeue
#SBATCH --output=/scratch/users/kalyanna/GM12878_microc/sbatch_logs/

source activate hic

resolutions=(250 500 1000 2000 4000 8000 16000 32000 64000)
genome="/scratch/users/kalyanna/ref/hg38.genome"
pairs="/scratch/users/kalyanna/GM12878_micro_c/mapped_pairs_MicroC_8
cools="/scratch/users/kalyanna/GM12878_micro_c/cools"
npz="/scratch/users/kalyanna/GM12878_micro_c/npz"

for res in "${resolutions[@]"; do
    echo "Processing at $res resolution"

```

```
cooler cload pairix -p 16 ${genome}:${res} $pairs $cools/GM12878.  
mkdir -p $npz/npz_${res}  
echo "Converting to npz"  
python /scratch/users/kalyanna/C.Origami/src/corigami/preprocess  
$cools/GM12878_microc_${res}.cool \  
$npz/npz_${res} \  
-r $res \  
--no-balance  
done
```

Final results

Npz matrices are stored at : `/scratch/users/kalyanna/GM12878_micro_c/npz/npz_${res}`

Cool files are stored at: `/scratch/users/kalyanna/GM12878_micro_c/cools`